

A 1-dimensional Convolutional Autoencoder (1D-CAE) assisted ‘feature learning’ for damage diagnosis in CFRP plate using Lamb waves.

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ABSTRACT

Standard machine learning algorithms traditionally address isolated tasks and require customization every time for a new kind of dataset. Thus, such a machine learning-based structural health monitoring approach lacks flexibility for any alteration in real-time field application. Most of the published articles have neither demonstrated nor discuss such capabilities. The present study attempts to overcome this by developing an Autoencoder based two-step diagnostic approach; In the first step, a 1D-CNN stacked auto-encoder (1D-CAE) is developed and trained to learn and extract the dominating features present in the Lamb waves. In the second step, the classification module is constructed, which inputs scaled features from the encoder and performs the classification between pristine and damaged conditions. The approach requires relatively minor training datasets than the conventional deep learning counterparts. The 1D-CAE is capable of auto-tuning, offers flexibility with different sizes of the 1D-input dataset. The study utilizes publicly available benchmark datasets collected by “Open guided waves”, which comprises Lamb wave signals recorded for a range of frequencies and artificial defects. The setup consists of arrays of transducers bonded to the $500\text{mm} \times 500\text{mm}$ and 2mm thin CFRP plate in “Pitch-Catch” configuration, which excites five cycles modulated Hann-filtered sine wave signal with an amplitude of $\pm 100\text{ V}$. Both the 1D-CAE and the classifier module is trained under the supervision of Adam optimization algorithm with varying learning rate. The fully trained classifier can detect the damage in the thin CFRP plate with 95% accuracy. In the later stage, the performance is evaluated against the unseen samples with defects generated from the experimental setup. The proposed algorithm achieves a high level of generalization.

Keywords: Structural health monitoring, Lamb waves, composite plate, autoencoder, 1D-CNN

1. INTRODUCTION

Evolving from Non-Destructive Testing (NDT), Structural Health Monitoring (SHM) is now recognized as a powerful tool for on-board condition monitoring and damage diagnosis. Research in SHM is in high demand for the aerospace industry, as these technologies can provide the best availability of aircraft. Among many SHM systems, the Lamb wave-based SHM system has become a well-established monitoring system, mainly because they are sensitive to smaller cracks and requires a relatively small number of sensors to scan a larger area [1][2][3]. However, analyzing the recorded Lamb waveform is often a challenge, as the modes generated due to damage is often lost or merged with the irrelevant artefacts coming from discontinuities like edges/boundaries, stiffeners, stringers et cetera [4]. The loss of damage related signatures from the recorded waveform can significantly reduce the efficiency of a diagnosing system. Recently, 1D and 2D convolutional neural network (CNN) based deep learning (DL) has been successfully used to develop data-driven intelligent Lamb wave-based SHM systems [5][6][7]. The CNNs are able to auto-extract higher-order damage related features, thus could provide a robust algorithm to analyze the raw signals directly [8][9]. CNN based SHM systems have so far been applied to metal and composite plate structures, however, this also reveals the issue of data dependence of such models. For each new structure, these DL models require a sufficient number of training samples to achieve the desired level of performance. Collecting a large number of samples is sometimes expensive, time-consuming and labour intensive.

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Such data dependence may limit the use of such DL-based SHM systems to their fullest extent. Thus, the aim of this study is to develop a two-stage deep learning-based SHM model for a $500\text{mm} \times 500\text{mm}$, 2mm thick CFRP composite panel using Lamb waves. In the first stage, 1D-CNN stacked autoencoder obtains the required features from raw Lamb signals. In the second stage, the encoder derived features are correlated to final state of the composite structure using a 1D-CNN based classifier. The study uses “Open Guided Waves” (OGW), a benchmark diagnostic data repository for training and evaluation of the proposed algorithm[10]. The OGW setup consists of arrays of transducers bonded to the CFRP plate in “Pitch-Catch” configuration, which excites five cycles modulated Hann-filtered sine wave signal with an amplitude of $\pm 100\text{ V}$.

The rest of the article is organised as follows. Section 2 describes the construction of the database from the OGW repository. Section 3 provides details on the development of the *Two-stage Autoencoder assisted 1D-CNN Classifier* model. The model performance characteristics are discussed in section 4. Finally, Section 5 summarizes the findings of the research.

2. CONSTRUCTION OF LAMB WAVE DATABASE

The current study uses OGW, a benchmark online data repository, consisting of recorded Lamb signals for a range of frequencies and artificial defects. These Lamb waves are experimentally generated from a CFRP composite panel of $500\text{mm} \times 500\text{mm}$ and 2mm thickness using a pair of six piezoelectric transducers (PZTs) bonded to the surface of the structure in a *pitch-catch* configuration, as shown in Fig. 1. Artificial defects have

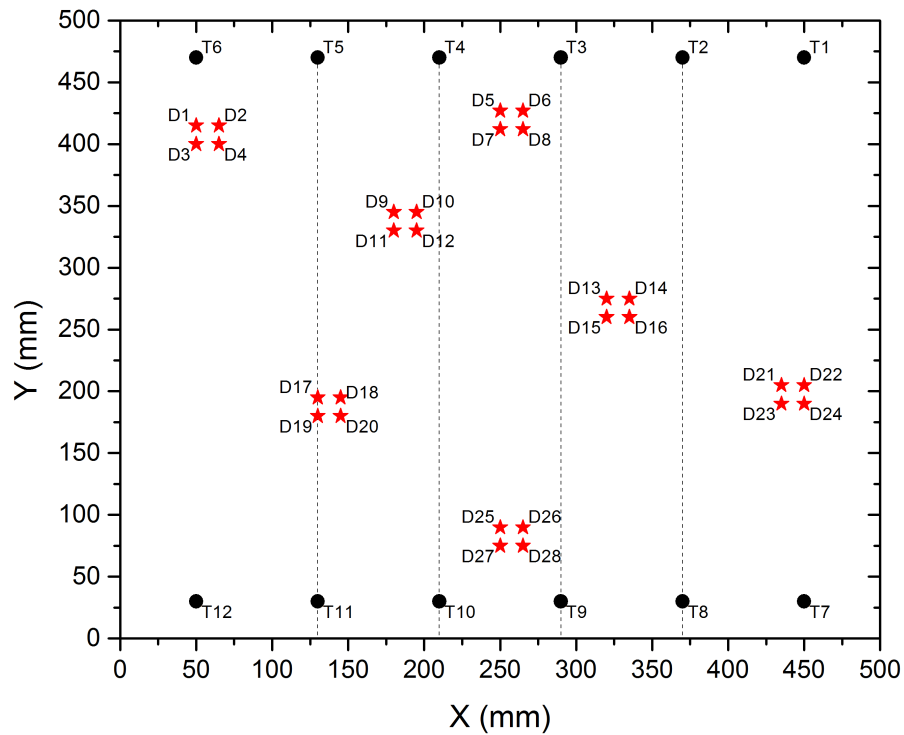


Figure 1. Piezoelectric transducer configuration with artificial defect location marked as D1-D28.

been realized by attaching an aluminium disc of mass 0.5 gm at twenty-eight different locations marked from D1 to D28, as shown in Fig. 1. The artificial damage position is distributed all across the CFRP plate. To scan the structure, a five cycle modulated Hann-filtered sine wave signal of amplitude $\pm 100\text{ V}$ is selected. The frequency of the scanning wave also varied in twelve steps from 40kHz to 260kHz. Originally, the entire plate structure is scanned in a round-robin manner, generating a total of sixty-six unique Lamb responses from twelve PZTs, where, each response consists of 13108 data-points. In order to reduce the computational cost, a simple

zone-wise scanning scheme is adopted. Here the entire plate area is divided into five rectangular zones, with each zone having four PZTs. With the zone-wise scanning scheme, the entire plate can be scanned with only thirty responses, where each zone generates a total of six responses.

2.1 Signal processing and database construction

As suggested in [10], all recorded signals are passed through a Butterworth high-pass filter. To amplify the damage signatures, differential signals are generated by subtracting the baseline signals from the recorded signals. Next, the resultant differential signals are down sampled by the factor of 3, to 4370 from 13108 data points.

To build the database, an equal number of damaged and pristine samples are arranged alternatively. Since the total number of damage scenarios is 28, an equal number of pristine samples are considered. Thus, the total strength of the database reaches $12 \times (28 + 28) = 672$ samples, where each sample is a matrix of size 4370×6 . From here, each sample is assigned a *sample ID* and the *condition* (i.e, the state of the structure, 0 for pristine and 1 for Damaged). Thus, the total size of each sample is now 4370×8 . Next, samples from the entire database are regrouped into a training database (D_{train}) and a testing database (D_{test}). D_{train} has 528 samples, while D_{test} has 144 samples in total.

3. DEEP LEARNING MODEL

The proposed deep learning model has two parts - the first part is a convolutional encoder, derived from a 1D-convolutional autoencoder and the second is a 1D-CNN based classifier, as shown in Fig. 3. Each part is briefly described as:

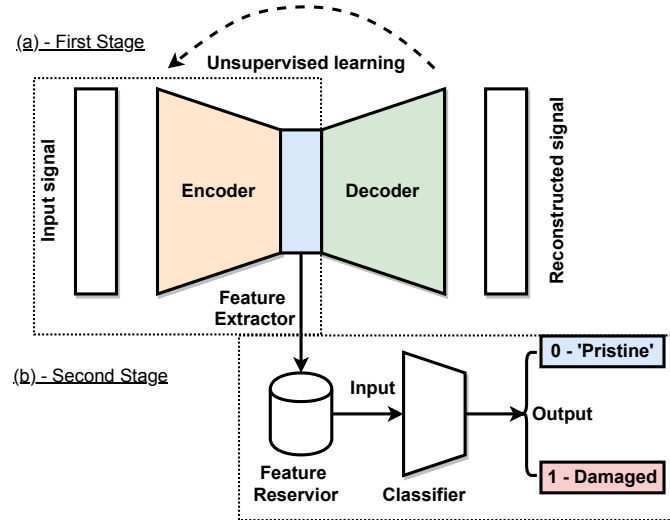


Figure 2. Workflow diagram of adopted methodology.

1. **1D-Convolutional Autoencoder (1D-CAE)** - The performance of any machine learning or deep learning model depends on the meaningfulness of the input features. In recent studies, the CNN-based autoencoder has proven to be particularly effective in enhancing the meaningful features from sensor data, as well as suppressing the irrelevant artefacts. Furthermore, the architecture of an autoencoder enables it to extract a compressed representation of the input signal from the output of the encoder. In the present research, the adopted 1D-CAE architecture has four 1D-convolutional layers and 1D-convolutional transpose layers in both the encoder and decoder units, respectively, while the feature extractor has only one 1D-convolutional layer, as shown in the Fig. 3. Each convolutional layer is followed by a LeakyReLU activation layer. However, the use of intermediate pooling layers is restricted, which avoids excessive compression and retains maximum spatial information at the 'bottleneck' region (i.e, feature extractor) of the 1D-CAE. The output of each layer is summarized in Table 1

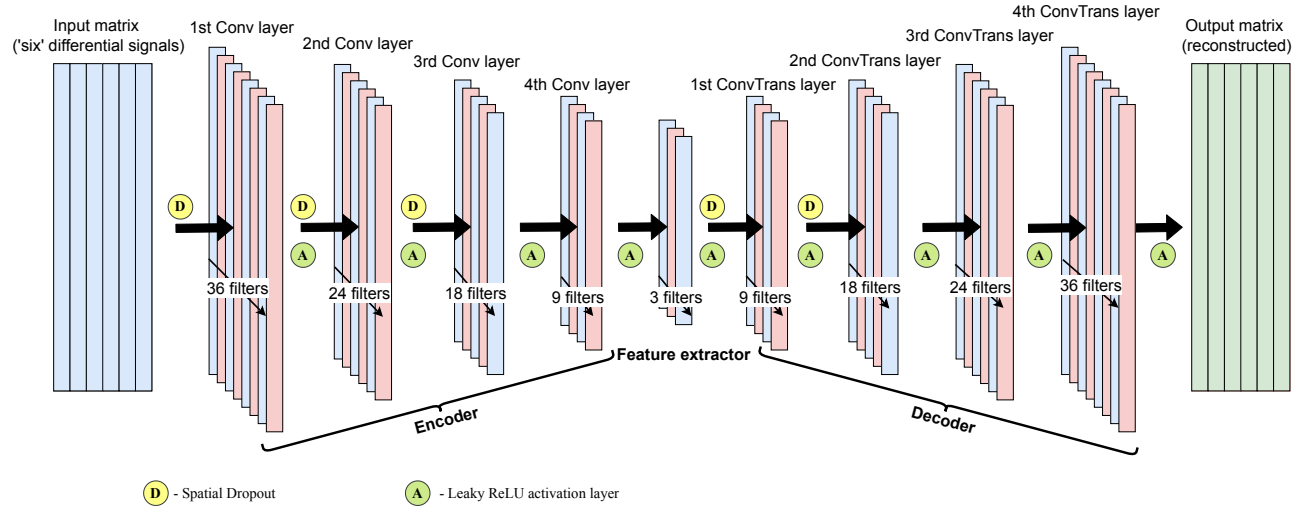


Figure 3. Schematic diagram of 1D-CAE configuration.

Table 1. The configuration of 1D convolutional stacked autoencoder.

Layers	Input shape	Output shape	Kernel size	Activation function
Input	$1 \times 4370 \times 6$	-	-	-
Conv1D	$1 \times 4370 \times 6$	$1 \times 4368 \times 36$	3	ReLU
Conv 1D	$1 \times 4368 \times 36$	$1 \times 4366 \times 24$	3	ReLU
Conv 1D	$1 \times 4366 \times 24$	$1 \times 4364 \times 18$	3	ReLU
Conv 1D	$1 \times 4364 \times 18$	$1 \times 4362 \times 9$	3	ReLU
Conv 1D	$1 \times 4362 \times 9$	$1 \times 4362 \times 3$	3	ReLU
ConvTrans 1D	$1 \times 4362 \times 3$	$1 \times 4362 \times 9$	3	ReLU
ConvTrans 1D	$1 \times 4362 \times 9$	$1 \times 4364 \times 18$	3	ReLU
ConvTrans 1D	$1 \times 4364 \times 18$	$1 \times 4366 \times 24$	3	ReLU
ConvTrans 1D	$1 \times 4366 \times 24$	$1 \times 4368 \times 36$	3	ReLU
Dense	$1 \times 4368 \times 36$	$1 \times 4370 \times 6$	6 (nodes)	ReLU

The output of the 1D-convolutional layer and the 1D-convolutional transpose layer is calculated as per Eqns. (1) and (2) respectively.

$$Output\ size = \left\lceil \frac{input\ size + 2 \times padding - (kernel\ size - 1) - 1}{stride} + 1 \right\rceil \quad (1)$$

$$Output\ size = (input\ size - 1) \times stride - 2 \times padding + kernel\ size \quad (2)$$

The Adaptive Moment Estimation (Adam) algorithm [11] is employed as the optimization algorithm, and the Huber loss function is deployed to update the weight matrices. Spatial Dropouts and L_2 kernel regularizer were imposed to avoid over-fitting while training of stacked auto-encoder. The batch size for training has been set to 32. A Reduce learning rate criterion is used, which is activated when the metric stops improving [50]. In the current model, once learning gets stagnant, the learning rate is reduced by a factor of 0.2. Hence the new learning rate will be -

$$New\ learning\ rate = learning\ rate \times factor$$

This callback monitors a loss metric and if the model is not showing any improvement for a 'five' consecutive epochs, then the learning rate is reduced. With these settings, the adopted unsupervised 1D-CAE model with a total of 9,957 trainable parameters is then trained for 500 epochs. Only 90% of the total samples

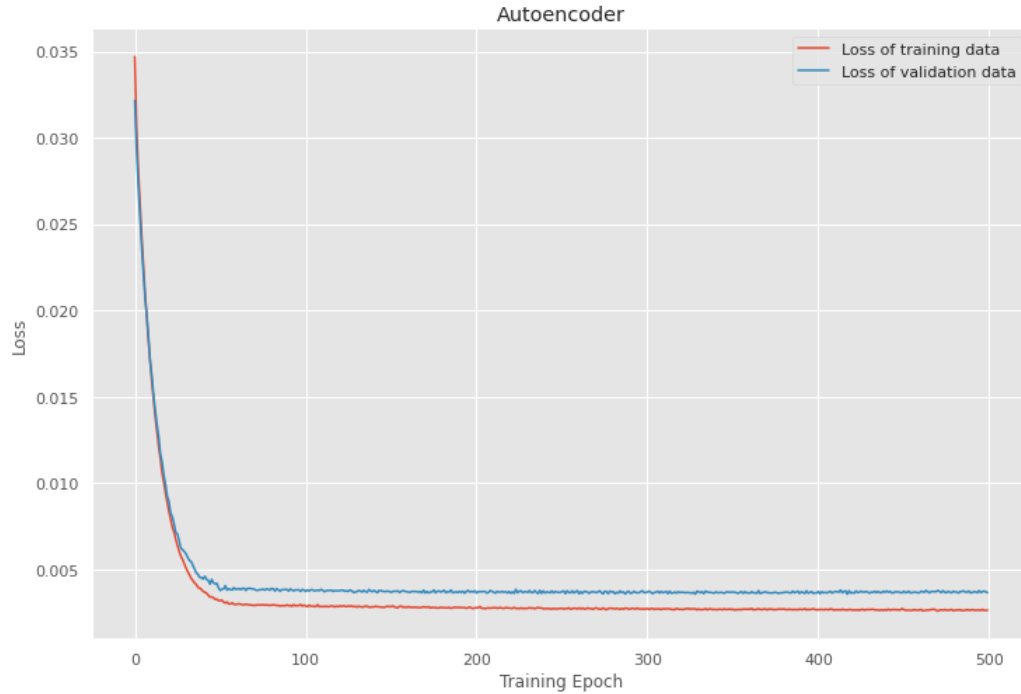


Figure 4. Training and validation loss of 1D-CAE(version - 1) versus number of epochs.

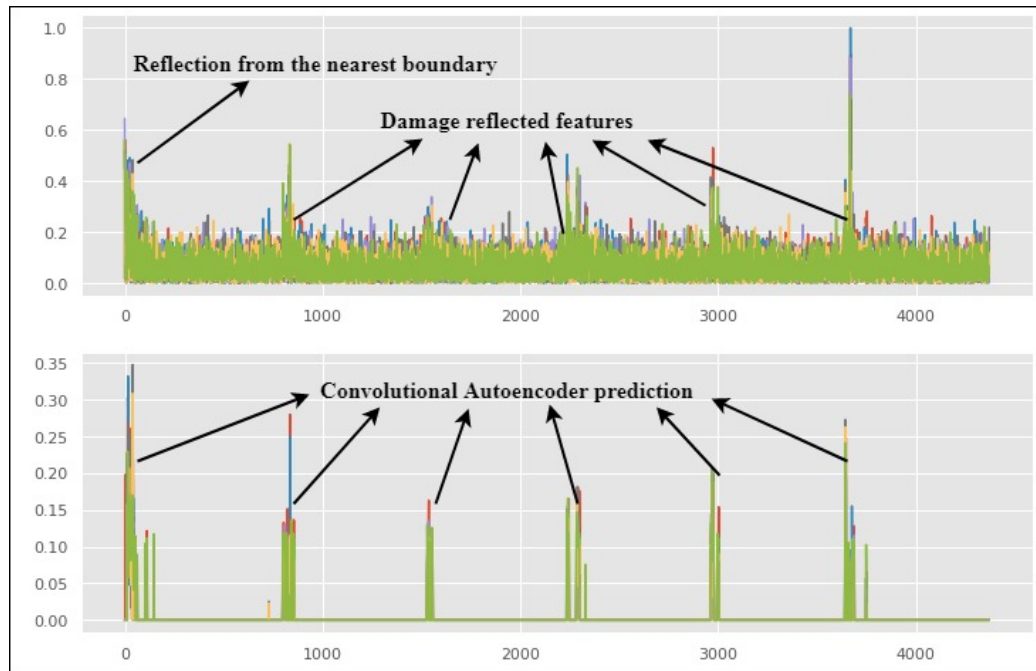
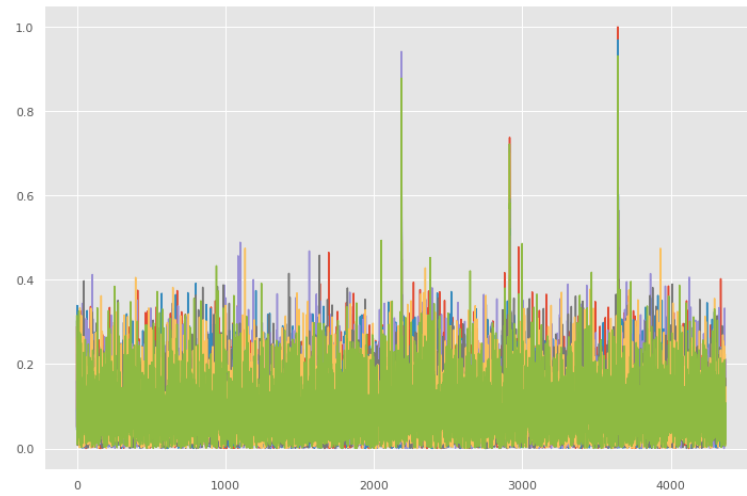


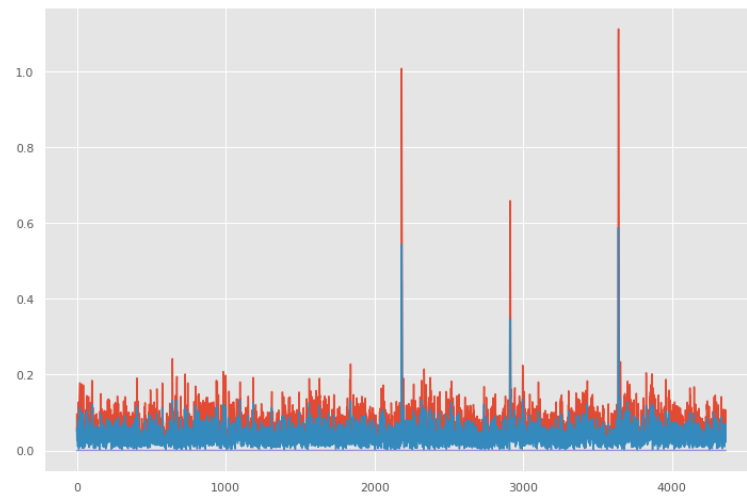
Figure 5. (Top) original Input sensor data from pristine sample, (bottom) 1D-CAE reconstructed data.

in the source database is utilized for training, and the remaining 10% is reserved for validation. Figure 4 presents the history of 1D-CAE's losses scored per epochs. This autoencoder can reproduce the input signals within a range of 10-12% of the reconstruction error, the reconstructed signal can be seen in Figure 5. The encoder unit is capable of suppressing the noise as well as amplifying key features from the input, which is quite evident from Figure 6. The set of features extracted by the encoder unit, are stored in

matrix format under a separate database named ‘feature reservoir’.



(a) Zone-5 with D-1 case scanned at 40 kHz.



(b) Encoder's output.

Figure 6. (a) - D1 sample at 40 kHz; (b) - Feature generated by encoder for D1 sample.

2. **1D-CNN based classifier** - In the second stage, the stored dominating features are used to train a shallow 1D-CNN classifier. The classifier module has a shallow configuration and consists of three layers. First layer is a 1D-CNN layer with kernel size equal to 9 along with three filters and uses 'ReLU' activation function with stride = 2, followed by max-pooling layer with a pooling window size = 5 and stride = 2. The output from max-pooling layer is flattened and submitted to the next layer. The second layer is a dense layer, also known as FCN or fully connected layer with 16 numbers of hidden neurons and uses 'ReLU' activation function. Finally, third and the last layer is also a Dense layer which does the classification between pristine (i.e., 0) and damage (i.e., 1) condition. Sigmoid activation function is used in binary classification layer. Classifier assigned binary cross-entropy as the loss function.

4. PERFORMANCE AND EVALUATION

Before training, the model hyper-parameters such as the batch size is set to 84, and the number of epochs is set to 2000. The classifier is trained using the Adam optimization algorithm with a learning rate is set at 0.00001. The total trainable parameters for the adopted classifier architecture is 52,310. Only 90% (i.e., 476) of the total samples in the training database is used to carry out training of the adopted model, while, the remaining 10% for validation of the model. Figure 7 presents the accuracy and losses history curve when the 1D-CNN classifier

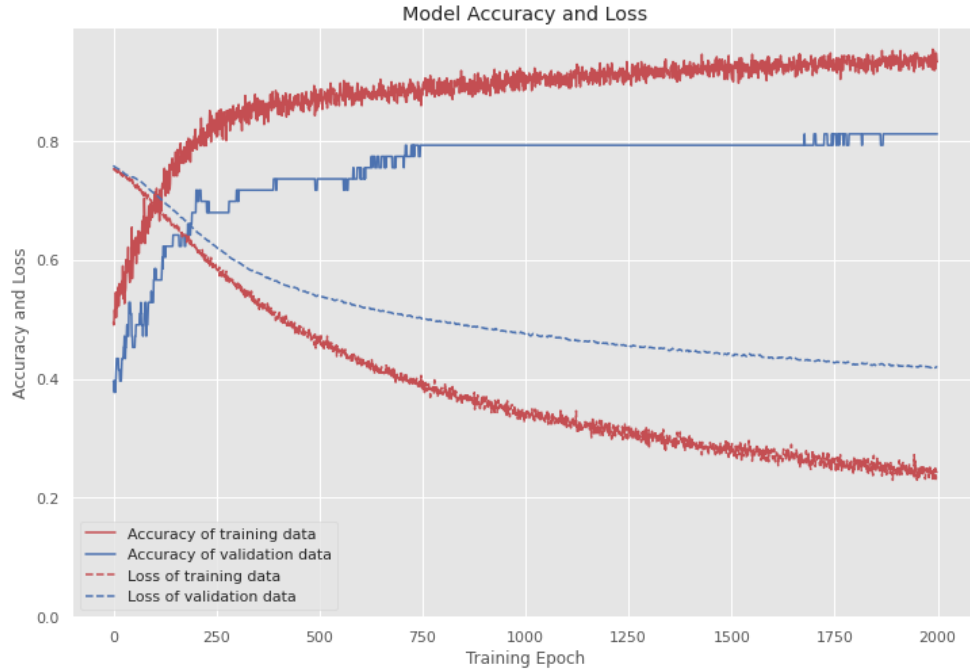


Figure 7. Training of 1D-CNN classifier using Adam optimization algorithm.

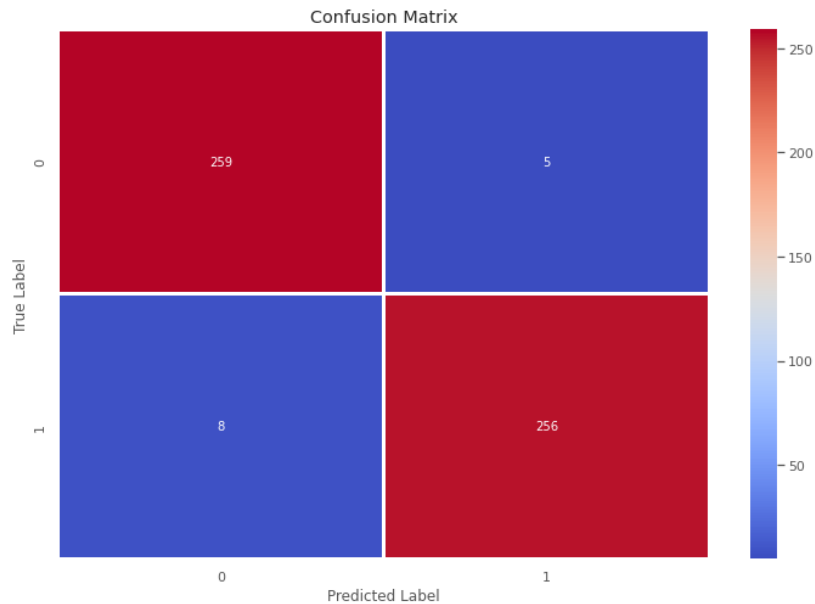


Figure 8. Classifier's confusion matrix for combined training and validation databases ($D_{T-train}$).

is trained with the Adam optimization technique. The Figure 8 represents the confusion matrix for 1D-CNN classifier, with an overall accuracy of 97.54% evaluated using the Equation 3.

$$Accuracy = \frac{TN + TP}{TN + TP + FN + FP} \quad (3)$$

Where, the first and fourth elements are described as the number of true-negative (TN) and true-positive (TP) predictions, and the second and third elements indicate the number of false-positive (FP) and false-negative (FN) predictions respectively. As, the model has achieved 97.54% accuracy in identify damage features. However, results based upon unseen datasets reveals the true classification capability of any classifier model. As shown in

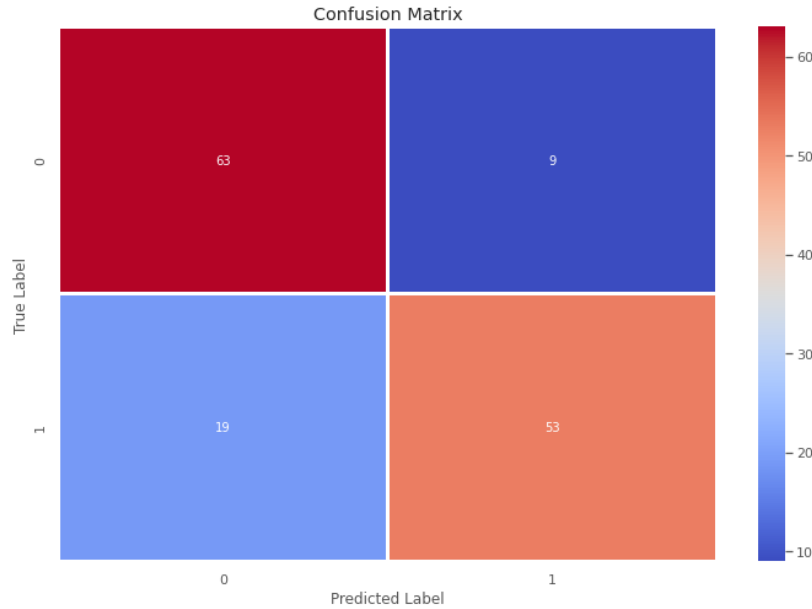


Figure 9. Classifier's confusion matrix for unseen test database (D_{test}).

Figure 9, out of 144 samples in the D_{test} database, classifier correctly predicted the condition of 116 samples, thus using Equation 3, the real-world classification accuracy is evaluated to be 80.5%. This proves that the autoencoder assisted classifier has achieved generalization over the problem.

5. CONCLUSION

The study present an implementation of a novel two-stage DL-based damage diagnosis of CFRP plate using Lamb waves. This approach has two stages: feature learning using 1D convolutional stacked auto-encoder, followed by 1D-CNN classifier for damage detection. The adopted approach offers modular architecture and capable to deliver robust classification performance with low computational cost. The strategy can be scaled up to scan and inspect the complete plate structure in a zone-wise or grid-wise manner. Such approach can predict multiple damages with minimal training support. Variations due to environmental factors like humidity and temperature are not included in the present investigation.

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